
The Anatomy of Market Stress: Decomposing, Explaining, and Acting on Financial Turbulence

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Abstract

Financial markets alternate between calm and stress, and risk models built on stable correlations react late when the switch comes. We measure market stress with one transparent number — the turbulence index of Kritzman and Li, computed daily on ten factor ETFs over two decades — and ask three questions of it. **Can it be read?** Yes. The index splits exactly into named factor contributions, so every crisis acquires a recognisable fingerprint: the 2018 “Volmageddon,” historically a blow-up of volatility products, is attributed almost entirely to the volatility factor without the method being told anything about it. Volatility also emerges as the clear leader of stress build-ups — a lead that survives rebuilding the index without the volatility factor — and a spillover analysis shows the whole system tightening around the dollar in stressed periods: flight-to-safety as a measured quantity rather than a story. **Can the next crisis be anticipated?** Partly, and this is the report’s new result. The factor mix of a coming stress episode is already visible in the weeks before it begins, far beyond what chance allows (a similarity of 0.72 where shuffled pairings produce 0.46), and it genuinely sharpens toward the eventual episode rather than merely persisting — a distinction we test three separate ways. Too few independent crises have occurred since 2020 to settle the sharpening claim out of sample, and we say so plainly. **Can it be traded?** No. Timing equity corrections is indistinguishable from a coin flip, and a pre-registered de-risking rule cuts the worst drawdown by roughly a fifth yet does no better than simply holding less equity throughout — protection comes from exposure, not foresight. Every result is produced with training strictly before testing, with the COVID crash inside the test window, and under an automated check that injects a fake future crash and verifies that no earlier number changes. The conclusion fits in one sentence: a stress signal is worth most as an interpreter — which factors, which type of crisis, which hedge class — and least as a timer.

1 The question, and the answer

Markets move between calm and stress, and the standard risk toolkit — built on stable correlations — reacts late when the switch happens. This project asks what a simple, fully transparent stress measure can and cannot deliver, and answers with one sentence that every result below either supports or bounds:

What a crisis will be made of is forecastable. When it hits is not.

Three questions organise the work, each a hypothesis stated in words: **(1) Read** — can one stress number be split into named factor contributions, and what leads its build-up? **(2) Anticipate** — do

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crises come in types, and is the mix of the *next* one visible in advance, beyond mere market inertia? **(3) Trade** — does any of this beat passive benchmarks out of sample, on risk-adjusted terms? The first two answers are yes (the second with a stated out-of-sample limit). The third is no — and the “no” is measured carefully enough to be a finding rather than a failure.

Hypotheses, stated so they can fail. **Visibility:** the run-up’s factor mix resembles its own coming episode far better than shuffled pairings allow, in and out of sample. **Sharpening:** the run-up improves on the episode’s own mix from a month earlier, under tests that respect how few independent crises there are. **Timing:** turbulence dynamics predict equity corrections well enough to beat simple persistence and passive benchmarks on risk-adjusted terms. The evidence establishes visibility in full; establishes sharpening in-sample while out of sample it is directional only, with a pooled multi-market test pre-stated as its decider; and finds no support for timing. One statement of power keeps the last claim honest: with roughly 70 independent monthly blocks and a Sortino resolution floor of ± 0.5 , this sample cannot *rule out* a small timing edge — what it establishes is that any such edge is smaller than two decades of data can resolve.

Where this sits in the literature. The turbulence lineage is the frame: Kritzman–Li [1] define the index; Kinlaw–Turkington [2] split it into magnitude and correlation surprise; Czaronis, Kritzman and Turkington add event time [3] and relevance-based prediction [4]. The Two Sigma Factor Lens [5] supplies the coordinate system, and its mixture-model regime detector [6] — which is explicitly non-predictive and collapses all acute stress into one “Crisis” state — is our supervised benchmark. Spillover measures [17, 7] provide the network apparatus; the realised-semivariance literature [8, 9, 10] motivates (and is contradicted by) our signed decomposition. Nearby literatures forecast something stress-related but not composition: stress *levels* [2], systemic-risk contributions [11], macro crisis classes [12]; systemic-stress composites [13] and the absorption ratio [14] monitor contemporaneously; regime-switching [15] and change-point [16] models date regimes without typing them. A documented search (Google Scholar, SSRN, RePEc, plus forward-citation tracing of the lineage papers) located no published work that predicts the *composition* — as opposed to the occurrence or level — of future market stress; we phrase the novelty as “unaware of any published work,” since unpublished proprietary research cannot be searched.

Contributions. (1) A leak-free, interpretable account of turbulence: an exact per-factor split used systematically to build a crisis-fingerprint catalogue and a lead–lag analysis that survives an obvious circularity check. (2) The principal novel result: the factor mix of a coming stress episode is visible in its run-up, robust to a battery of adversarial controls, with the sharpening component honestly bounded out of sample. (3) A disciplined set of negative results delimiting what can be traded, capped by a pre-registered de-risking test against the correct static benchmark. Every claim is reported as a distribution rather than a point, and traces to a numbered, reproducible pipeline.

2 Data, and three honesty rules

Data. Ten factors of the Two Sigma Factor Lens [5], each proxied by one liquid ETF: equity (SPY), rates (TLT), credit (HYG), commodities (GSG), emerging markets (EEM), the dollar (UUP), inflation (TIP), value (IWD), quality (QUAL), and volatility ($\sqrt{\text{VIX}}$). Daily prices 2005–2026, all public. We use ten factors rather than the fuller fourteen because the extra four only have data from about 2011: a fourteen-factor version cannot see the 2008 and 2011 crises — the two most informative episodes — and we verified it is also no better out of sample (AUCs within the noise band on every model).

Rule 1 — nothing sees the future, and code proves it. Every statistic uses only data available the day before. This is not a promise: an automated check injects an artificial extreme value on a chosen date and verifies, byte for byte, that no number before that date changes. Over 200 such checks run with the test suite. We adopted this discipline after auditing an earlier version of the pipeline and finding 48 look-ahead defects; that version was discarded and everything rebuilt. Model training follows the same logic: training folds always precede test folds, with a 21-day gap so no forward-looking label straddles the boundary [20], and predicted probabilities are calibrated on training data only.

Rule 2 — COVID sits inside the test set. Models are built on data through 2019 and tested on 2020–2026 ($\approx 1,580$ trading days), a split fixed in advance. Any claim that survives had to survive the hardest stress event of the modern sample without having seen it.

Rule 3 — distributions, not lucky numbers. A backtest depends on where the train/test boundary falls, so we slide that boundary across 49 positions and report the whole distribution. This rule has teeth: an earlier configuration reported a $+0.055$ Sortino-ratio edge (return per unit of downside risk) that turned out to be the single best of the 49 boundaries — the median was -0.087 . We disclosed it and dropped it. Resampling month-long blocks of the test window (a block bootstrap, which respects the fact that daily returns are not independent) shows Sortino differences below about ± 0.5 cannot be resolved on two decades of data, so we never claim them.

3 One number for market stress

The turbulence index of Kritzman and Li [1] asks, each day: how unusual is today’s *joint* move across the ten factors, given history?

$$d_t = (\mathbf{y}_t - \boldsymbol{\mu})^\top \Omega^{-1} (\mathbf{y}_t - \boldsymbol{\mu}), \quad (1)$$

where $\mathbf{y}_t$ is today’s factor returns and $\boldsymbol{\mu}, \Omega$ are the mean and covariance from a rolling ten-year window, stabilised with Ledoit–Wolf shrinkage [19] (a standard, parameter-free conditioning of the covariance). This is the Mahalanobis distance — the standard measure of how far a point sits from a distribution’s centre, in correlation-aware units. The index flags both big moves and *strange* moves: factors that usually move together suddenly diverging counts as stress even when each move is small. It is one number, from one window, with nothing fitted — which is why it can be audited. Alternatives were considered and rejected on the same criterion: rolling volatility sees only magnitude; principal-component distances lose the named factor axes; mixture models and neural detectors fit latent states that cannot be audited. Turbulence is the one measure that is simultaneously correlation-aware, economically named, and exactly decomposable.

It behaves as it should (Figure 1): every named crisis of the sample (2008, 2011, 2015, both 2018 episodes, 2020, 2022) sits in its top percentiles, and stressed days are persistent and clustered. One striking property, replicating Czaronis, Kritzman and Turkington [3] on independent data: measured per unit of *accumulated stress* rather than per calendar day, the famous fat tails of equity returns largely disappear — excess kurtosis, the standard measure of how much heavier a distribution’s tails are than the normal bell curve’s, falls 94%, from 8.81 to 0.57 (Figure 2). Stress-time, not calendar time, is the clock on which markets are approximately normal.

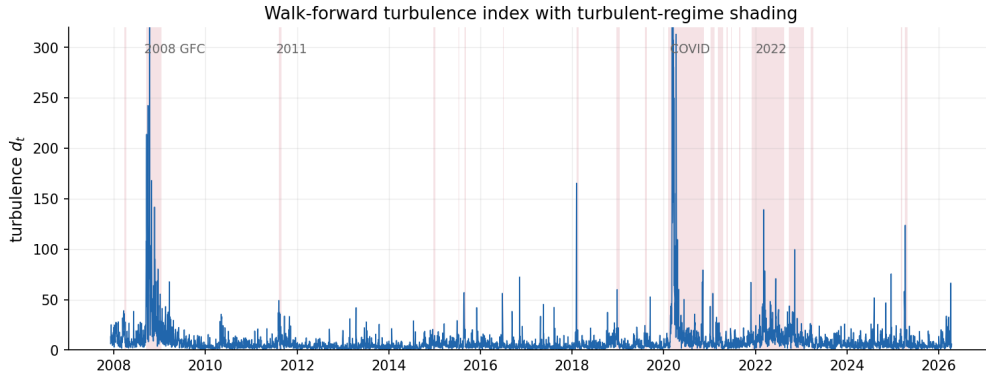


Figure 1: The walk-forward turbulence index (2007–2026) with stress shading (top decile of trailing turbulence). Every named modern crisis is recovered; the series is strictly causal — statistics through $t - 1$ only — and byte-invariant under injected future shocks.

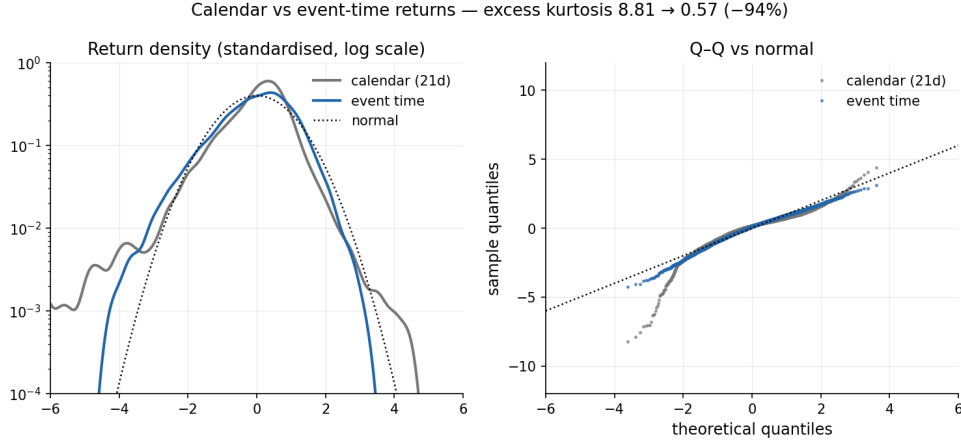


Figure 2: Equity returns per calendar day versus per unit of accumulated stress. On the stress clock the heavy tails largely vanish: excess kurtosis 8.81 → 0.57 (Jarque–Bera normality statistic 17,204 → 293).

4 Reading the number

4.1 Every crisis has a fingerprint

The index splits exactly: factor i ’s contribution $c_i = \delta_i(\Omega^{-1}\delta)_i$ sums back to the whole to machine precision ($\sim 10^{-16}$), using the same rolling windows — so the attribution is as leak-free as the index itself. Averaging contributions over a crisis gives its *fingerprint* (Figure 3, Table 1).

The fingerprints match the economic record without being told anything about it. The clearest case: February 2018 (“Volmageddon”), historically a blow-up of volatility products, comes out 79% attributable to the volatility factor. Three reading rules travel with the table. First, the ETFs are correlated with each other, so shares are attributions, not exact variance splits — the *ranking* of drivers is the reliable object, and rankings are stable when the window length changes (volatility ranks first in Volmageddon at 5-, 10- and 15-year windows; the share moves 58/79/80%). Second, the earliest fingerprints (2008, 2011) rest partly on spliced pre-inception proxies and shorter covariance windows, so they are indicative rather than exact. Third, the 2020 row surprises people: in factor space the COVID crash was above all a *value* dislocation with credit second, because equity’s move — though enormous — was typical of how equity moves in a sell-off, and the index rewards unusual co-movement, not size.

Table 1: Crisis fingerprints: leading factor contributions in each named window.

Crisis	1st	2nd	3rd
2008 financial crisis	credit 23%	equity 21%	value 15%
2011 euro / US downgrade	rates 36%	volatility 21%	value 10%
2015 China / oil	volatility 27%	commodities 16%	dollar 15%
2018 Volmageddon	volatility 79%	equity 7%	commodities 5%
2018 Q4 sell-off	equity 57%	quality 22%	commodities 13%
2020 COVID	value 31%	credit 18%	quality 15%
2022 inflation shock	quality 21%	equity 17%	commodities 15%

4.2 What leads the build-up

For each factor we ask Granger’s question [18], plainly: does the factor’s recent past improve the prediction of tomorrow’s *change* in stress, beyond what stress’s own past already tells you? On several thousand daily observations almost everything clears conventional significance, so the informative object is the *strength ranking*, measured as each factor’s incremental R^2 — the extra share of next-day variation it explains. Volatility leads decisively (incremental $R^2 = 1.5\%$, roughly twice the next group: quality 0.8%, equity 0.8%, credit 0.7%, value 0.7%); the slow macro variables — dollar, inflation — explain essentially nothing. The obvious objection is circularity: volatility sits *inside* the

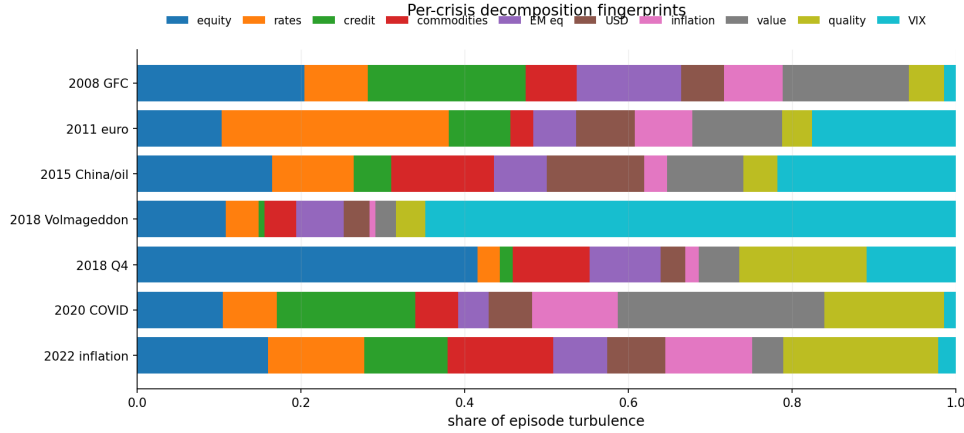


Figure 3: The stress index decomposed into factor contributions over time. Each crisis is dominated by a distinct, recognisable set of drivers.

index. So we rebuilt the index on the nine non-volatility factors: volatility still leads the volatility-free index, decisively ($p = 7 \times 10^{-14}$).

4.3 Under stress, the system sinks into the dollar

How the factors *transmit* stress to each other is measured with the spillover decomposition of Diebold and Yilmaz [17]: fit a vector autoregression — a regression of each factor’s return on everyone’s recent past — and then ask what share of each factor’s forecast-error variance comes from shocks to the others rather than to itself. Estimated separately on calm and turbulent days (Figure 4), two things happen under stress. Total connectedness — the average share of variance that is cross-factor — rises from 61% to 65%: the system tightens. And the dollar’s *net* position (variance transmitted minus received) deepens from -17% to -25% , making it the largest net receiver in the stressed system, while equity, value and quality are the main transmitters. This is flight-to-safety not as a story but as a measured quantity — and it quantifies the correlation surge that the Factor Lens paper records but does not model. Throughout, these lead-lag relations are read as statistical precedence, not as interventions.

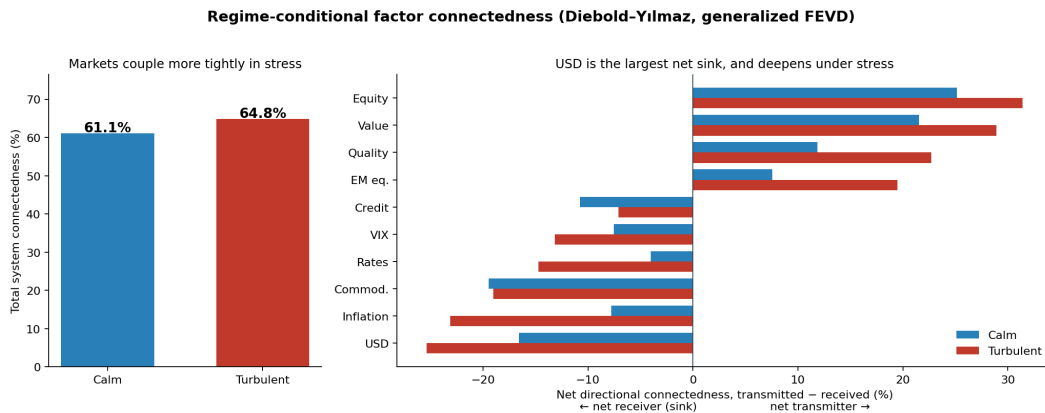


Figure 4: Diebold–Yilmaz spillovers, calm versus turbulent days (turbulent = top decile of the index). Left: total connectedness rises under stress. Right: net directional position by factor — the dollar is the largest net receiver of variance, and deepens under stress.

5 The main result: the next crisis’s mix is visible in advance

Setup. A stress *episode* begins when yesterday’s turbulence crosses its own trailing 90th percentile, with nearby stressed days merged: 38 episodes between 2009 and 2026, median length nine days. (2008 cannot appear as an episode — by the time the rolling threshold exists, 2008-level stress *is* the trailing norm; its fingerprint above uses a named calendar window instead.) Grouping the 38 fingerprints (Ward’s clustering, an agglomerative method that repeatedly merges the two most similar groups, with the number of groups chosen by fit rather than by hand) gives three recurring types — volatility-led (11 episodes), equity-led (7), and value-and-commodities-led (20) — moderately stable under resampling (an adjusted-Rand agreement of 0.68 with the full-sample grouping, on a scale where 1 means identical groupings and 0 means chance). Where the industry-standard regime model [6] sees one undifferentiated “Crisis” state, the decomposition sees three (Figure 5).

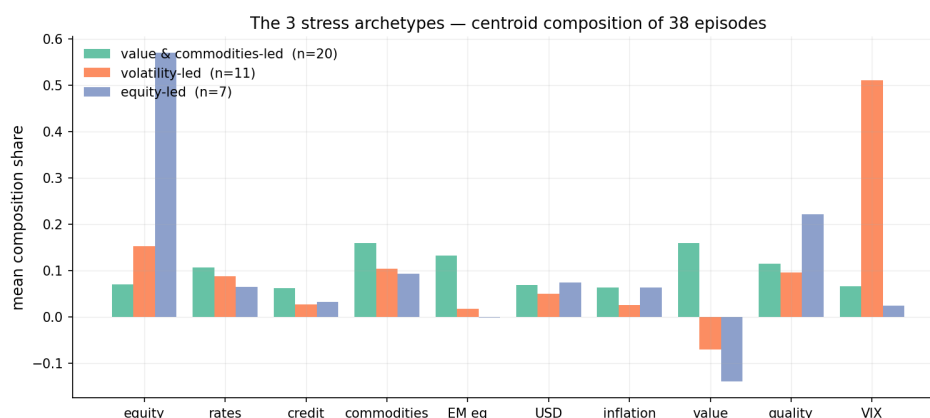


Figure 5: The 38 high-turbulence episodes grouped by composition fingerprint into three recurring types (Ward’s clustering; number of groups chosen by fit). Resampling agreement 0.68: stable at the core, softer at the boundaries.

The test. For each episode, take the average factor mix over the 21 days *before* it begins (a horizon fixed in advance, never tuned) and compare it with the mix of the episode itself, using the angle between the two mixes (cosine similarity). Chance comes from a *shuffle test* (a permutation test): scramble which run-up goes with which episode 10,000 times and see where the real similarity falls among the scrambled ones. The shuffle is the right benchmark because it preserves everything about the individual mixes — their scale, their concentration, their typical shape — and destroys only the *pairing*. Whatever similarity survives shuffling is what run-ups share with episodes in general; the excess above it is what a run-up knows about its *own* episode in particular.

Result 1 — the mix is visible. Real similarity **0.715** against a shuffled level of **0.463**, $p < 10^{-4}$. This is not an artifact of the measure or of the moving covariance window: it is unchanged under rank-based and overlap-based similarity measures, unchanged in log-ratio geometry, and unchanged when every episode is recomputed under one fixed covariance matrix (0.715 vs. 0.474) — the full battery is one line each in Appendix A. It holds on the 2020–2026 test years alone (0.698 vs. 0.475, $p < 10^{-4}$), and re-running the entire pipeline on a second universe — nine US sector ETFs — reproduces it (0.777 vs. 0.615; Figure 7). One caution stated plainly: sectors co-move with the equity factor, so the second universe bounds the concern that this is a Factor-Lens artifact; it is not independent confirmation.

Result 2 — the run-up genuinely sharpens toward the episode. Visibility alone is not yet anticipation, and the distinction deserves care. Markets are persistent: a stressed week tends to be followed by another stressed week, so a run-up could resemble its episode simply because the market was *already* in that shape — inertia, not foresight. Worse, our episodes are defined by rising turbulence, so the days just before an episode are rising-stress days by construction, and some resemblance is built in. Before claiming anticipation, we therefore have to show that the run-up carries information beyond what the market’s recent past already contained. Three checks do this, each aimed at a different way the claim could be an illusion.

First, against the episode's own past. If the run-up were pure inertia, the market's mix from a month earlier — before the build-up began — should predict the episode just as well as the final three weeks do. So for every episode we run that exact horse race, pairing each run-up against the same episode's own earlier self (a paired comparison, which needs no assumption about episodes resembling one another). The run-up wins, by +0.107 in similarity ($p = 0.003$, 95% interval $[+0.043, +0.173]$), and the picture behind the number is a steady climb: measured four months out, three months out, two, then one, similarity to the eventual episode rises monotonically, $0.54 \rightarrow 0.58 \rightarrow 0.61 \rightarrow 0.72$ (Figure 6). The market does not sit in a fixed crisis shape; it converges on one.

Second, counting crises honestly. A statistical test is only as good as its count of independent observations, and our 38 episodes are not 38 independent draws — the successive legs of 2020, or of 2022, share one macro backdrop and one unfolding story. Treating them separately would quietly flatter the p-value. We therefore group episodes separated by less than ninety days into single crises — 38 episodes become roughly 14 — and re-test the climb with just one observation per crisis, the most conservative counting this sample allows. The climb survives: $p = 0.022$, with a 95% interval on the convergence slope of $[-0.104, -0.017]$, excluding zero.

Third, against a placebo. There is a subtler trap: any quantity measured *backwards* from a chosen endpoint tends to drift toward that endpoint, because nearby windows are always more correlated with it than distant ones. A rising climb could therefore be an artifact of hindsight measurement rather than a property of crises. The clean control is to run the identical measurement from 400 randomly chosen calm dates — moments when nothing was about to happen. If the climb were a measurement artifact, it would appear there at full strength. It appears at only a third of the strength (a slope of -0.019 against the real episodes' -0.056 ; $p = 0.006$): the sharpening belongs to crises, not to the ruler.

Finally, because similarity scores can feel generous — two mixes can score well by agreeing on the unimportant factors — we also pose the question in its bluntest form: can the run-up simply *name* the single factor that will dominate the coming episode? It can, 39% of the time (33% on the test years), where a guesser who knows only how often each factor tends to dominate would manage 18%. Roughly twice chance, with no similarity measure involved at all.

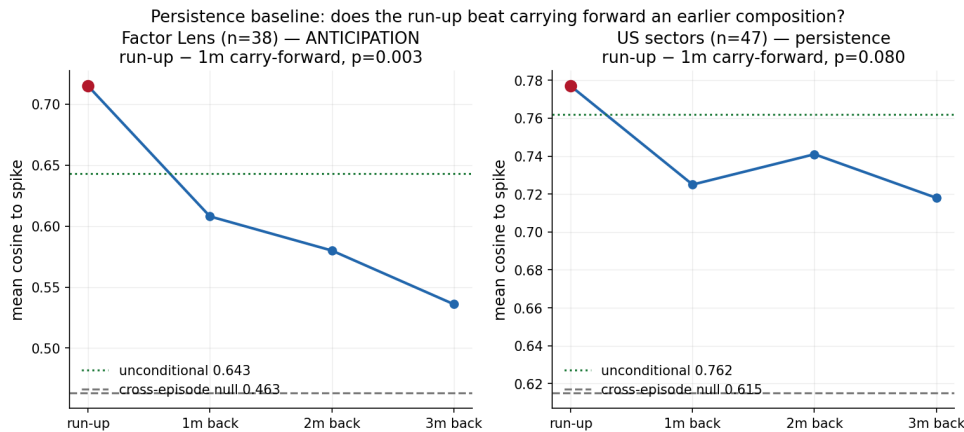


Figure 6: The run-up's similarity to the eventual episode, measured one to four months out. It beats every earlier-window baseline and climbs monotonically as the episode approaches — a climb that survives crisis-level testing ($p = 0.022$) and a random-date placebo ($p = 0.006$).

The honest limit. On the test years alone the sharpening is positive (+0.102) but not decisive: only about eight independent crises have occurred since 2020, and the resampled 95% interval, $[-0.019, +0.202]$, includes zero. The claim therefore splits. *That the coming mix is visible* is established, in and out of sample. *That it sharpens beyond inertia* is established on the full sample and suggestive out of sample. One experiment settles it — pooling episodes across many markets to raise the number of independent crises — and we name it as the decider rather than argue past the data.

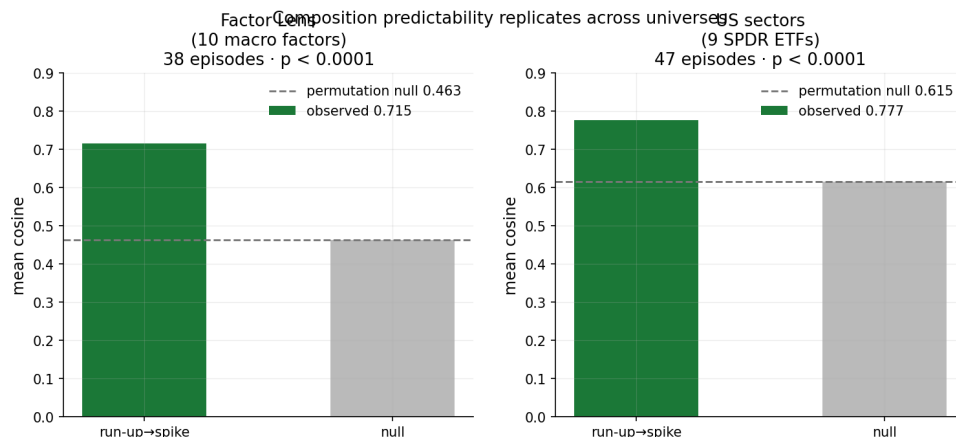


Figure 7: The same test on two universes: ten macro factors (left) and nine US sector ETFs (right). Visibility of the coming mix clears the shuffle level on both ($p < 10^{-4}$).

6 Can you trade it?

6.1 Timing: no

The trading question is asked with deliberately simple machinery. Four standard classifiers — two regularised logistic regressions, a linear support-vector machine, and a random forest, spanning simple linear rules and a non-linear ensemble — are given features that are nothing more than readings of the turbulence index itself: today’s level, its averages over the past week and month, how quickly it has been rising, and how long the market has been in the stressed regime. No outside data enters, so the test isolates one question: what does turbulence alone know about the future? The models are fixed once and never tuned per experiment — with so few crises in the sample, tuning would let luck masquerade as skill. Discrimination is scored by AUC: the probability that the model ranks a randomly chosen crisis day above a randomly chosen calm day, so 0.5 is a coin flip and 1.0 is perfect.

Two forecasting targets must be kept apart, because they behave completely differently. The first — *will the equity market fall at least 5% in the coming month?* — is the question a trader wants answered, and the answer is chance: AUC 0.482, with a median of 0.49 and a spread of [0.45, 0.54] across all 49 train/test boundaries. The second — *will the stress index itself rise?* — looks far better at first sight: AUC 0.71. But that number needs the right yardstick. Turbulence is a slow-moving series — stressed weeks follow stressed weeks, and in our stress-heavy test window 86% of days are already “stressed or about to be” — so a forecaster who simply says *tomorrow will look like today* scores 0.63 with no model at all. Once uncertainty is measured honestly (by resampling month-long blocks of the test window, since daily outcomes are not independent), the model’s interval [0.62, 0.81] contains that no-model floor. The honest reading: stress predicts stress mostly because stress persists, and the model adds little beyond the persistence. One genuinely useful property survives, and it is worth naming precisely: *calibration*. A forecast is calibrated when its probabilities mean what they say — of the days the model calls “30% risk,” about 30% should actually turn bad. Mapping raw model scores into calibrated probabilities (learned on training data only) cuts the Brier score — the mean squared error of probability forecasts, lower is better — from 0.343 to 0.156. Even where the model cannot *rank* days well, its probability statements are honest, which is exactly what a risk report needs.

The timing null is not for lack of trying: we attacked it from four directions, each chosen because the literature or common sense suggested it should help.

Direction. The volatility literature holds that *downside* moves carry the predictive content — “bad volatility” forecasts, “good volatility” does not. We tested whether that carries over by splitting every factor’s turbulence contribution, exactly, by the direction of the underlying move. It does not carry over: the loss side and the gain side are equally informative (AUC 0.692 vs. 0.693). Whatever is true of a single asset’s volatility, *direction* adds nothing to a cross-market unusualness measure.

The stress clock. Since returns are near-normal when measured in stress-time (Section 3), one might hope forecasts improve on that clock too. They do not: the stress-clock forecast is a slightly weaker classifier (AUC 0.48 $\rightarrow$ 0.44, partly because the stress-dense test years compress one “event” to roughly 14 calendar days), though the *evaluation* does become more stable (positive at 43% of boundaries versus 20%). Further variations — event-clock features, per-market clocks — move nothing material either (they are catalogued in the technical companion). The clock changes the statistics of the problem, not its forecastability.

The correlation channel. We tested directly the published claim [2] that the *correlation* part of turbulence — how unusual the co-movement is, as opposed to how large the moves are — carries extra warning content for future volatility. On this panel it does not: days of unusual correlation are followed by only marginally higher volatility (26.4% vs. 24.1% annualised, never significant), and the correlation part adds zero incremental R^2 once simple persistence and the magnitude part (which adds 3.6 points) are accounted for.

The benchmark detector. Finally, perhaps a different regime representation would do better. We rebuilt the industry-standard mixture-model detector [6] cleanly and gave both methods identical supervised tasks — a comparison fair by design, since the two feature sets share every non-regime column and differ only in how the regime is represented. Turbulence wins on point AUC for all four classifiers (gaps +0.07 to +0.19), but a paired test on the two models’ forecast errors (Diebold–Mariano, on the Brier score) confirms the gap for the random forest only ($p = 0.001$): a win on point discrimination, confirmed on one model of four — not a sweep.

One positive: grade the severity, don’t flag it. One intervention does move the needle, and it deserves a closer look because of what it teaches about representation. The baseline strategy sees stress through a *binary* flag — the market is either “stressed” (turbulence above its 90th percentile) or it is not. A binary flag forces an all-or-nothing response: the model can shift its position by one fixed amount when the flag switches on, and by nothing otherwise, so a mild wobble and a full-blown crisis trigger the same reaction. The intervention replaces the flag with an *ordinal severity scale*: turbulence is graded into ordered levels (calm, elevated, high, severe, extreme) by its position in the trailing distribution, computed causally like everything else. The same linear model can now respond *proportionally* — trim a little at “elevated,” cut hard at “extreme.”

The effect is the sharpest positive in the trading section (Figure 8): the linear models’ median Sortino lift across the 49 train/test boundaries flips from -0.15 with the binary flag to $+0.15$ with the five-level scale, positive at 90% of boundaries — and the improvement is not specific to the number five, holding at three, five and ten levels alike. Two honest qualifications frame it. The random forest does *not* benefit (its median worsens to -0.26) — which is itself informative, since a tree can already carve its own thresholds from the raw turbulence level; the gain belongs to models that cannot bend, which is exactly what the representation story predicts. And because the intervention was found by searching representations rather than specified in advance, we grade it exploratory. The message survives both caveats: severity is information that a yes/no flag throws away, and *how you represent* stress exposure is improvable even where *timing* is not — a theme the calibrated, graded probability of the next subsection picks up directly.

6.2 Drawdown protection: yes — but from exposure, not foresight

If the *when* cannot be forecast, the practical question becomes the *what*: given that stress is arriving, which asset protects? The answer depends on the type of crisis, for a concrete economic reason. In a flight-to-safety crisis (COVID), investors rush into government debt, rates fall, and long Treasuries rally; in an inflation or rates shock (2022), rates rise and those same Treasuries sit at the epicentre of the loss. Table 2 shows the flip in the two held-out crises: +14.9% for bonds through COVID, -31.2% through 2022, with gold roughly flat in the latter. No single blind hedge is safe across both types.

This is where the decomposition earns a practical role — but only its *signed* version. The magnitude of the rates contribution cannot choose the hedge: a large rates contribution says only that rates are moving unusually, not *which way*, and both crisis types load heavily on rates. The signed decomposition, which splits each factor’s contribution by the direction of the underlying move, does know the difference between a rates rally and a rates crash. A hedge routed on that signal never sits on the wrong side of the flip, and cuts buy-and-hold’s worst drawdown from -33.7% to -25.1% .

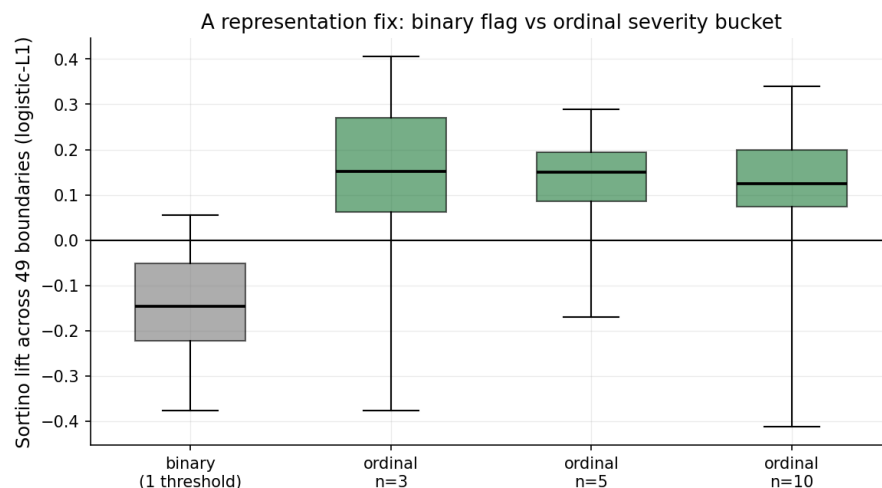


Figure 8: Grading severity beats flagging it. Distribution of the risk-adjusted (Sortino) lift across the 49 train/test boundaries for the linear model: the binary stressed/calm flag (grey) is negative at the median, while ordinal severity scales (green) are positive — at three, five, and ten levels alike, so the gain is not an artifact of one chosen granularity.

Table 2: The working hedge flips with the kind of stress (held-out episodes; total instrument return over the episode).

Episode	stress type	bonds (TLT)	gold (GLD)	equity (SPY)
2020 COVID (Feb–Apr)	flight to safety	+14.9%	+5.2%	−13.2%
2022 inflation (year)	rates shock	−31.2%	−0.8%	−18.2%

Honesty requires two qualifications. First, the routed hedge is statistically indistinguishable from a naive static 50/50 bond–gold mix — diversification alone achieves most of the protection. Second, a control that isolates exposure — a passive position scaled down to the overlay’s realised volatility, about 81% equity — shows both sides of the coin: the vol-matched position draws down *deeper* (−29.8% vs. −27.9%), so the reduction is not just lower exposure, yet it earns a far better Sortino ratio (0.80 vs. 0.32), so the overlay pays for its protection in risk-adjusted return. The demonstrated value is diversification across crisis types and a shallower worst drawdown — not clever routing, and not a better return per unit of risk.

After this many negative results, one worry deserves to be closed properly: perhaps some combination of choices — a different threshold, a friendlier benchmark, a kinder metric — would have shown a timing edge, and we simply had not looked. The remedy is *pre-registration*: fix the target, the trading rule, the benchmark and the verdict criterion *before* running the experiment, so that no outcome, good or bad, can be the product of searching. The design reads in one breath. Build a calibrated probability that the market falls in the coming month, from the turbulence level and its one-month change and nothing else; move the portfolio to cash whenever that probability crosses a threshold; freeze the threshold on 2010–2014 data and never touch it again; test on 2015–2026, which contains both COVID and the 2022 shock; sweep the definition of a “fall” across 3%, 5% and 8% so the headline cannot be one lucky cut; and — the decisive choice — compare not only against buy-and-hold but against **a static portfolio holding the rule’s average equity exposure throughout**. The static benchmark is what isolates timing: it holds the same average amount of equity and differs only in *when* the equity is held. If the rule cannot beat it, the rule’s protection comes from holding less, not from knowing when.

The results (Table 3, Figure 9) look excellent at first: worst drawdown cut from 33.7% to 26.5%, Sortino 1.23 against 0.97, Calmar — return per unit of worst drawdown — 0.53 against 0.40. But the comparison that matters is the last row of the table. Against the static benchmark, the Calmar difference has a resampled 95% interval of [−0.21, +0.66]: it spans zero, meaning the data cannot tell the rule apart from its timing-free twin. And the apparent edge exists only at the 5% threshold —

at 3% and 8% the two are flat ties. The verdict we committed to in advance therefore applies: **the protection comes from holding less equity, not from knowing when.**

Table 3: The pre-registered de-risking rule, test period 2015–2026 (5% threshold). “Static” holds the rule’s average equity exposure constantly — the benchmark that isolates timing.

	De-risking rule	Buy & hold	Static
Worst drawdown	26.5%	33.7%	32.9%
Sortino ratio	1.23	0.97	0.97
Calmar ratio	0.53	0.40	0.40
Δ Calmar vs. static (95% interval)	[−0.21, +0.66] — spans zero		

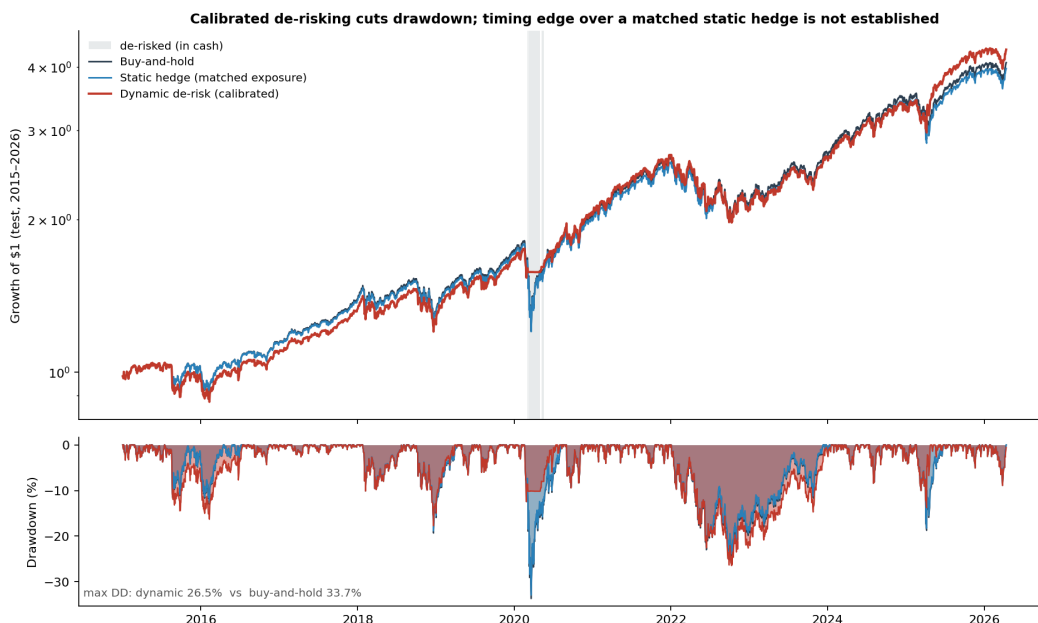


Figure 9: The pre-registered rule (red) against buy-and-hold (dark) and the matched-exposure static portfolio (blue), 2015–2026. Top: growth of \$1; grey bands mark days in cash. Bottom: drawdowns — the rule shallows COVID and 2022 but tracks the static portfolio too closely for a timing claim.

7 Parameter sensitivity: what moves, what does not

Every load-bearing parameter was swept, on the principle that sensitivity analysis is how one learns a model’s behaviour (Table 4). Two lessons generalise. First, *rankings are far more stable than shares or point estimates*: driver rankings, fingerprint orderings and the three-type structure survive window and threshold changes that move the underlying numbers materially — which is why this report states its claims at the rank and type level wherever possible. Second, where a result *is* parameter-sensitive — the de-risking edge, which appears only at the 5% threshold — that sensitivity is itself reported as the finding.

8 What this adds, and what it cannot say

Table 5 grades every finding by how solid it is; the rest of this section says what the collection amounts to.

What it adds. A transparent stress measure that can be read (fingerprints, leading factors, measured spillovers); a genuinely new observation — the factor mix of a coming stress episode is visible before it starts; and a disciplined account of what cannot be done, capped by a pre-registered test of the de-risking question.

Table 4: Parameter-sensitivity inventory. All sweeps are walk-forward and leak-free.

Parameter	Sweep	Behaviour learned
Covariance window	5 / 10 / 15y	fingerprint <i>rankings</i> stable (volatility first in Volmageddon at every window: 58/79/80%); 10y best on entry AUC
Stress threshold	85 / 90 / 95th pct	episode structure robust
Factor set	10 vs. 14	the 14-factor series starts 2013 — blind to 2008/2011 — and is no better out of sample; coverage decides
Covariance shrinkage	measured path	≤ 0.018 throughout, smallest in concentrated crises — not driving fingerprints
Run-up length	fixed 21d in advance	never searched, so the flagship test carries no selection bias
De-risk stress threshold	3 / 5 / 8%	apparent timing edge only at 5% — reported as a tie, per the pre-registered rule
Transaction costs	5 bps per switch	conclusions unchanged

Table 5: Every finding, graded.

Finding	How solid
Stress-time removes most fat tails (−94% kurtosis)	Rock-solid (replication)
No look-ahead, proven by code (200+ checks)	Rock-solid
Every crisis has a recognisable fingerprint	Strong
Volatility leads the build-up (survives the no-VIX rebuild)	Strong
System tightens and sinks into the dollar under stress	Strong (descriptive)
The coming crisis’s mix is visible in advance	Strong; new; holds out of sample
... and sharpens beyond inertia	Strong in-sample; suggestive out of sample
Beats the rebuilt mixture-model detector on point AUC	Moderate (confirmed on 1 of 4 models)
De-risking cuts worst drawdown (33.7% → 26.5%)	Real; no timing skill vs. holding less
Timing; direction; stress-clock timing; correlation-surprise content	No — clean, informative negatives

What it cannot say. Everything predictive rests on few independent crises (roughly fourteen in total, eight in the test years) — the binding constraint on every claim; the sector replication shares the equity factor, so it bounds rather than confirms; and the ETF basis is correlated, so fingerprint shares are attributions, not variance splits. None of this is fixable by better modelling — only by more crises.

The one next experiment. Pool episodes across many markets and forecast the coming mix out of sample at scale. It either upgrades the sharpening result from suggestive to established, or honestly retires it. Both outcomes are worth having.

Reproducibility. Every number, table and figure comes from a single numbered pipeline on public data (`python run_all.py -core` for this report’s numbers); each step writes a plain-English report; the no-look-ahead check runs with the test suite.

A The robustness checks

Check	Outcome
Alternative similarity measures	Rank-based 0.493 vs. 0.138; overlap-based 0.640 vs. 0.485; log-ratio geometry 0.445 vs. 0.150 — all $p < 10^{-4}$
Fixed covariance matrix for all episodes	0.715 vs. 0.474 — unchanged; not a moving-window artifact
Covariance window 5 / 10 / 15 years	Driver rankings stable; Volmageddon volatility-led at every window (58/79/80%)
Regime threshold 85th / 90th / 95th percentile	Episode structure robust
Ten vs. fourteen factors	Fourteen-factor series blind to 2008/2011 and no better — ten wins on coverage
Run-up length	Fixed at 21 days in advance; never searched
Covariance shrinkage intensity	≤ 0.018 throughout and smallest in the concentrated crises — not driving fingerprints
De-risk stress threshold 3% / 5% / 8%	Apparent timing edge only at 5% — reported as a tie, per the pre-registered rule
Transaction costs (5 bps per switch)	Conclusions unchanged

References

- [1] Kritzman, M. & Li, Y. (2010). Skulls, financial turbulence, and risk management. *Financial Analysts Journal*, 66(5):30–41.
- [2] Kinlaw, W. & Turkington, D. (2014). Correlation surprise. *Journal of Asset Management*, 14(6):385–399.
- [3] Czaronis, M., Kritzman, M. & Turkington, D. (2023). Event time. *The Journal of Portfolio Management*, 49(4).
- [4] Czaronis, M., Kritzman, M. & Turkington, D. (2024). Relevance-based prediction: a transparent and adaptive alternative to machine learning. *The Journal of Financial Data Science*, 6(1).
- [5] Duncombe, G. & Kay, B. (2018). Introducing the Two Sigma Factor Lens. Two Sigma white paper.
- [6] Botte, A. & Bao, D. (2021). A machine learning approach to regime modeling. Two Sigma Street View.
- [7] Billio, M., Getmansky, M., Lo, A.W. & Pelizzon, L. (2012). Econometric measures of connectedness and systemic risk in the finance and insurance sectors. *Journal of Financial Economics*, 104(3):535–559.
- [8] Barndorff-Nielsen, O.E., Kinnebrock, S. & Shephard, N. (2010). Measuring downside risk: realised semivariance. In *Volatility and Time Series Econometrics: Essays in Honor of Robert F. Engle*. Oxford University Press.
- [9] Patton, A.J. & Sheppard, K. (2015). Good volatility, bad volatility: signed jumps and the persistence of volatility. *Review of Economics and Statistics*, 97(3):683–697.
- [10] Bollerslev, T., Li, J., Patton, A.J. & Quaadvlieg, R. (2020). Realized semicovariances. *Econometrica*, 88(4):1515–1551.
- [11] Adrian, T. & Brunnermeier, M.K. (2016). CoVaR. *American Economic Review*, 106(7):1705–1741.
- [12] Kaminsky, G.L. (2006). Currency crises: are they all the same? *Journal of International Money and Finance*, 25(3):503–527.
- [13] Holló, D., Kremer, M. & Lo Duca, M. (2012). CISS — a composite indicator of systemic stress in the financial system. ECB Working Paper 1426.
- [14] Kritzman, M., Li, Y., Page, S. & Rigobon, R. (2011). Principal components as a measure of systemic risk. *The Journal of Portfolio Management*, 37(4):112–126.
- [15] Hamilton, J.D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2):357–384.
- [16] Adams, R.P. & MacKay, D.J.C. (2007). Bayesian online changepoint detection. *arXiv:0710.3742*.
- [17] Diebold, F.X. & Yılmaz, K. (2014). On the network topology of variance decompositions: measuring the connectedness of financial firms. *Journal of Econometrics*, 182(1):119–134.
- [18] Granger, C.W.J. (1969). Investigating causal relations by econometric models and cross-spectral methods. *Econometrica*, 37(3):424–438.
- [19] Ledoit, O. & Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411.
- [20] López de Prado, M. (2018). *Advances in Financial Machine Learning*. Wiley.

AI Usage Statement

Claude (Anthropic) was used for: (1) **code assistance** — generating, debugging and optimising the Python pipeline; (2) **language editing** — proofreading and improving the clarity of written sections. All methodological decisions, result interpretations, and conclusions are the author’s own.